

A Lean-Verified Single-Transition Theorem for CMV Ball Density

Proof built, Lean-verified, and semantically accepted by
Agentic Lean Math Assistant · github.com/RosaRojacr/agentic-lean-math-assistant

The candidate comparison

Cañete, Miranda Jr., and Vittone study the planar isoperimetric problem with density

$$f_\lambda(x) = \begin{cases} \lambda, & |x| \leq 1, \\ 1, & |x| > 1, \end{cases} \quad 0 < \lambda < 1.$$

Weighted area is the integral of this density over a region; weighted perimeter uses the same density on its boundary. In particular, an arc on the unit circle has weight λ . The relevant candidates are regions of type B, bounded by an arc of the unit circle and an exterior circular arc meeting it at angle $\arccos\lambda$, and balls of type C, whose boundaries meet the unit circle orthogonally. The source classification and candidate formulas are in [CMV, Proposition 3.22, pp. 32–33, and Theorem 3.23, pp. 34–36]. We prove an analytic comparison of these two complete candidate families. Its application to arbitrary finite-perimeter minimizers requires the source-level geometric premises discussed at the end.

For a weighted area v , let $\mathcal{P}_B(v)$ and $\mathcal{P}_C(v)$ be the sets of weighted perimeters of all admissible candidates of the indicated type having area exactly v , and define

$$I_B(v) = \inf \mathcal{P}_B(v), \quad I_C(v) = \inf \mathcal{P}_C(v).$$

The admissible parameter domains and formulas are specified below. On the domain $v > \lambda\pi$ used here, both sets are nonempty singletons; thus these definitions neither select a favorable branch nor depend on the value assigned to an empty infimum.

LEAN REFERENCES

```
CMVBallCase.lean CMVBallCase.typeBPerimeterValuesAtArea;
                  CMVBallCase.typeCPerimeterValuesAtArea; CMVBallCase.typeBPerimeterAtArea;
                  CMVBallCase.typeCPerimeterAtArea; CMVBallCase.exists_typeBRadius_at_area;
                  CMVBallCase.exists_typeCRadius_at_area;
                  CMVBallCase.typeBPerimeterAtArea_eq_radius;
                  CMVBallCase.typeCPerimeterAtArea_eq_radius
```

Theorem. Let $0 < \lambda < 1$ and $\lambda\pi < v_0 < v$. If

$$I_C(v_0) \leq I_B(v_0),$$

then

$$I_C(v) < I_B(v).$$

Thus weak dominance of type C at one such area becomes strict dominance at every larger area.

This is the formal persistence theorem, not an unconditional classification of isoperimetric regions.

LEAN REFERENCES

```
CMVBallCase.lean CMVBallCase.typeC_dominance_persists
```

Candidate formulas and global parameters

Fix $0 < \lambda < 1$ throughout the proof, and put

$$a = \arccos \lambda, \quad s = \sqrt{1 - \lambda^2}, \quad S(t) = t - \sin t \cos t.$$

For a unit circle, $S(t)$ is the area of the circular segment with central angle $2t$, for $0 \leq t \leq \pi$. For type B, take

$$0 < \beta \leq \pi - a, \quad \alpha = \beta + a, \quad r = \frac{\sin \alpha}{\sin \beta}.$$

The two boundary arcs have lengths $2r(\pi - \beta)$ and 2α . The region contains the unit disk. Its Euclidean area is the sum of the two circular segments cut by their common chord; subtracting $(1 - \lambda)\pi$ gives its weighted area. Consequently the formulas in [CMV, p. 34, Figure 19 on p. 35] are

$$\begin{aligned} P_B(\beta) &= 2((\pi - \beta)r + \lambda\alpha), \\ A_B(\beta) &= r^2(\pi - \beta + \sin \beta \cos \beta) + S(\alpha) - (1 - \lambda)\pi. \end{aligned}$$

The endpoint $\beta = \pi - a$ is retained: it has $r = 0$ and represents the unit disk. The endpoint $\beta = 0$ printed in the source is an infinite-radius limit, not a finite candidate.

LEAN REFERENCES

```
CMVBallCase.lean CMVBallCase.BAdmissible; CMVBallCase.typeBRadius_endpoint_zero;
                  CMVBallCase.arcSegment
CMVBallDefinitions.lean CMVBallCase.typeBAlpha; CMVBallCase.typeBRadius;
                        CMVBallCase.typeBPerimeter; CMVBallCase.typeBArea
```

For type C, let $0 < b < \pi/2$, and set $R = \cot b$. The inner and outer boundary arcs have lengths $2Rb$ and $2R(\pi - b)$. Orthogonality gives an overlap with the unit disk consisting of segments of areas $R^2 S(b)$ and $S(\pi/2 - b)$. Hence [CMV, p. 35]

$$\begin{aligned} P_C(b) &= 2R(\pi - (1 - \lambda)b), \\ A_C(b) &= R^2(\pi - (1 - \lambda)S(b)) - (1 - \lambda)S(\pi/2 - b). \end{aligned}$$

These are exactly the source's formulas with $R = 1/\tan b$. Both angular endpoints are excluded degenerations. The elementary geometric descriptions explain the formulas; the formal objects are the explicit real-valued formulas, not sets equipped with a formalized perimeter measure.

LEAN REFERENCES

```
CMVBallCase.lean CMVBallCase.CAdmissible
CMVBallDefinitions.lean CMVBallCase.typeCRadius; CMVBallCase.typeCPerimeter;
                        CMVBallCase.typeCArea
```

Use radius as a global parameter. The identity and inverse charts are

$$r = \lambda + \operatorname{scot} \beta, \quad \beta_\lambda(R) = \frac{\pi}{2} - \arctan \frac{R - \lambda}{s}, \quad b(R) = \arctan \frac{1}{R}.$$

The first chart is a bijection between the finite type-B angle domain and $R \geq 0$; the second is a bijection between $0 < b < \pi/2$ and $R > 0$. The radius maps are strictly decreasing in their angle parameters, and the displayed inverses work on their entire domains. In particular, the type-B chart includes $R < \lambda$, where $\beta_\lambda(R) > \pi/2$, as well as $R \geq \lambda$; no quadrant is discarded.

LEAN REFERENCES

```
CMVBallCase.lean CMVBallCase.typeBRRadius_eq_add_sqrt_mul_cot; CMVBallCase.betaAtRadius;
                  CMVBallCase.bAtRadius; CMVBallCase.typeBRRadius_strictAntiOn;
                  CMVBallCase.typeCRadius_strictAntiOn; CMVBallCase.betaAtRadius_admissible;
                  CMVBallCase.typeBRRadius_betaAtRadius; CMVBallCase.betaAtRadius_typeBRRadius;
                  CMVBallCase.bAtRadius_admissible; CMVBallCase.typeCRadius_bAtRadius;
                  CMVBallCase.bAtRadius_typeCRadius
```

Write $P_B(R), A_B(R)$ for the type-B formulas evaluated at $\beta_\lambda(R)$, and $P_C(R), A_C(R)$ for the type-C formulas evaluated at $b(R)$. Thus

$$\begin{aligned} P_B(R) &= 2((\pi - \beta_\lambda(R))R + \lambda(\beta_\lambda(R) + a)), \\ A_B(R) &= R^2(\pi - S(\beta_\lambda(R))) + S(\beta_\lambda(R) + a) - (1 - \lambda)\pi, \\ P_C(R) &= 2R(\pi - (1 - \lambda)b(R)), \\ A_C(R) &= R^2(\pi - (1 - \lambda)S(b(R))) - (1 - \lambda)S(\pi/2 - b(R)). \end{aligned}$$

The radius formulas agree with the angular formulas for every admissible radius.

LEAN REFERENCES

```
CMVBallCase.lean CMVBallCase.typeBPerimeterAtRadius; CMVBallCase.typeBAreaAtRadius;
CMVBallCase.typeCPerimeterAtRadius; CMVBallCase.typeCAreaAtRadius;
CMVBallCase.typeBPerimeterAtRadius_agrees;
CMVBallCase.typeBAreaAtRadius_agrees;
CMVBallCase.typeCPerimeterAtRadius_agrees;
CMVBallCase.typeCAreaAtRadius_agrees
```

Area and perimeter monotonicity

Differentiation, using $S'(t) = 2\sin^2 t$, gives, for $R > 0$,

$$\begin{aligned} P_B'(R) &= 2(\pi - S(\beta_\lambda(R))), & A_B'(R) &= R P_B'(R), \\ P_C'(R) &= 2(\pi - (1 - \lambda)S(b(R))), & A_C'(R) &= R P_C'(R). \end{aligned}$$

Both perimeter derivatives are positive. Indeed, the first factor is $S(\pi - \beta_\lambda(R)) > 0$, while $0 < S(b(R)) < b(R) < \pi/2$ makes the second factor positive. Thus area and perimeter are strictly increasing in radius: on $[0, \infty)$ for type B, and on $(0, \infty)$ for type C. Continuity at the type-B endpoint supplies the closed endpoint assertion.

LEAN REFERENCES

```
CMVBallCase.lean CMVBallCase.arcSegment_hasDerivAt; CMVBallCase.arcSegment_pos;
CMVBallCase.arcSegment_lt_self;
CMVBallCase.typeBPerimeterAtRadius_hasDerivAt;
CMVBallCase.typeCPerimeterAtRadius_hasDerivAt;
CMVBallCase.typeBAreaAtRadius_hasDerivAt;
CMVBallCase.typeCAreaAtRadius_hasDerivAt;
CMVBallCase.typeB_perimeter_factor_pos;
CMVBallCase.typeC_perimeter_factor_pos;
CMVBallCase.typeBAreaAtRadius_strictMonoOn;
CMVBallCase.typeCAreaAtRadius_strictMonoOn;
CMVBallCase.typeBPerimeterAtRadius_strictMonoOn;
CMVBallCase.typeCPerimeterAtRadius_strictMonoOn
```

The relevant area ranges follow without an arbitrary inverse choice. For type B,

$$A_B(0) = \lambda\pi, \quad P_B(0) = 2\lambda\pi, \quad A_B(R) > \frac{\pi}{2}R^2 - \pi \quad (R \geq \lambda).$$

For type C, the useful bounds are

$$\lambda\pi R^2 - \frac{\pi}{2} < A_C(R) < \pi R^2 \quad (R > 0).$$

In particular, $A_C(\lambda) < \pi\lambda^2 < \lambda\pi$, and its lower bound is unbounded above. The intermediate value theorem and strict monotonicity therefore give a unique positive radius of each type at every area $v > \lambda\pi$. The inverse angle charts show that every admissible angular candidate of that area has this radius and this angle. Consequently the all-candidate perimeter sets are singletons and the infima are their actual perimeter values. Moreover, I_B is strictly increasing on $(\lambda\pi, \infty)$, and $I_B(v) > 2\lambda\pi$ there.

LEAN REFERENCES

```
CMVBallCase.lean CMVBallCase.typeBAreaAtRadius_zero; CMVBallCase.typeBPerimeterAtRadius_zero;
CMVBallCase.typeBAreaAtRadius_large_lower;
CMVBallCase.typeCAreaAtRadius_lt_full;
CMVBallCase.typeCAreaAtRadius_large_lower;
CMVBallCase.exists_typeBRadius_at_area;
CMVBallCase.exists_typeCRadius_at_area;
CMVBallCase.typeB_values_eq_singleton;
CMVBallCase.typeC_values_eq_singleton;
CMVBallCase.typeBPerimeterAtArea_eq_radius;
CMVBallCase.typeCPerimeterAtArea_eq_radius;
CMVBallCase.typeBPerimeterAtArea_strictMonoOn;
CMVBallCase.typeBPerimeterAtArea_gt_commonEndpoint
```

The fixed-radius comparison

We next prove the decisive inequality

$$P_C(R) < P_B(R) \quad (R > 0).$$

This is a comparison at equal radii, not at equal areas. Half the perimeter difference is

$$H(\lambda, R) = (\lambda - R)\beta_\lambda(R) + \lambda \arccos \lambda + (1 - \lambda)Rb(R).$$

For $0 < R \leq \lambda$, the first term is nonnegative and the other two are positive, so $H(\lambda, R) > 0$.

LEAN REFERENCES

```
CMVBallCase.lean CMVBallCase.fixedRadiusHalfGap;
CMVBallCase.perimeter_gap_eq_two_mul_halfGap;
CMVBallCase.fixedRadiusHalfGap_pos_of_le
```

For the remaining region $R > \lambda$, vary the density while keeping R fixed. For $0 < c < 1$, define

$$\begin{aligned} D(c, R) &= (R - c)^2 + 1 - c^2, \\ L(c, R) &= \beta_c(R) - b(R) - \frac{c\sqrt{1-c^2}}{D(c, R)}, \\ N(c, R) &= R^3 - 4R^2c + 2Rc^2 + 3R - 2c. \end{aligned}$$

Here $D(c, R) > 0$, and direct differentiation yields

$$\frac{\partial}{\partial c} \frac{H(c, R)}{c} = \frac{R}{c^2} L(c, R), \quad \frac{\partial L}{\partial c}(c, R) = -\frac{cN(c, R)}{\sqrt{1-c^2} D(c, R)^2}.$$

The polynomial has the particularly useful derivative and diagonal value

$$\frac{\partial N}{\partial c}(c, R) = -\left(4\left(R - \frac{1}{2}\right)^2 + 1 + 4R(1-c)\right) < 0 \quad (R > 0, c < 1), \quad N(R, R) = R(1 - R^2).$$

Thus $N(\cdot, R)$ is strictly decreasing on $(-\infty, 1]$.

LEAN REFERENCES

```
CMVBallCase.lean CMVBallCase.fixedRadiusD; CMVBallCase.fixedRadiusL; CMVBallCase.fixedRadiusN;
CMVBallCase.fixedRadiusD_pos;
CMVBallCase.fixedRadiusHalfGap_density_ratio_hasDerivAt;
CMVBallCase.fixedRadiusL_density_hasDerivAt;
CMVBallCase.fixedRadiusN_density_derivative_identity;
CMVBallCase.fixedRadiusN_density_derivative_neg;
CMVBallCase.fixedRadiusN_diagonal; CMVBallCase.fixedRadiusN_strictAntiOn
```

We claim $L(c, R) < 0$ for $0 < c < \min(1, R)$. First, $L(0, R) = 0$, and $L(\cdot, R)$ is continuous at zero. If $N(c, R) \geq 0$, then $N(t, R) > 0$ for $0 < t < c$. Hence L strictly decreases from zero on $[0, c]$, proving the claim, including the case $N(c, R) = 0$. If $R \leq 1$, this case always applies, because

$$N(c, R) > N(R, R) = R(1 - R^2) \geq 0.$$

If instead $N(c, R) < 0$, necessarily $R > 1$. For every $c < t < 1$, we have $N(t, R) < 0$, so L is strictly increasing to its left limit at one. The reciprocal angle formula

$$\beta_t(R) = \arctan \frac{\sqrt{1-t^2}}{R-t} \quad (t < R)$$

shows that this limit is $-b(R) < 0$. Therefore $L(c, R) \leq -b(R) < 0$ in this case as well. The restriction $c < R$ is essential to this argument.

LEAN REFERENCES

```
CMVBallCase.lean CMVBallCase.fixedRadiusL_zero; CMVBallCase.fixedRadiusL_continuousAt_zero;
CMVBallCase.fixedRadiusL_neg_of_N_nonneg;
CMVBallCase.fixedRadiusN_pos_of_radius_le_one;
CMVBallCase.betaAtRadius_eq_arctan_reciprocal;
CMVBallCase.fixedRadiusL_tendsto_one; CMVBallCase.fixedRadiusL_neg_of_N_neg;
CMVBallCase.fixedRadiusL_neg
```

It follows that $H(c, R)/c$ is strictly decreasing for $0 < c < \min(1, R)$. If $\lambda < R < 1$, continuity at $c = R$ and the already proved case of equal density and radius give

$$\frac{H(\lambda, R)}{\lambda} > \frac{H(R, R)}{R} > 0.$$

If $R \geq 1$, then

$$\lim_{c \uparrow 1} \frac{H(c, R)}{c} = 0.$$

For $R > 1$ this follows from the reciprocal angle formula. For $R = 1$, the bound $0 < \beta_c(1) \leq \pi/2$ makes $(1 - c)\beta_c(1) \rightarrow 0$, while the remaining terms tend to zero. Strict decrease and this limit imply $H(\lambda, R)/\lambda > 0$: choose an intermediate density $\lambda < c < 1$, whose ratio is nonnegative by the limit, and compare strictly with it. Together with the case $R \leq \lambda$, this proves the fixed-radius inequality for every $R > 0$.

LEAN REFERENCES

```
CMVBallCase.lean CMVBallCase.fixedRadiusHalfGap_density_ratio_strictAntiOn;
CMVBallCase.fixedRadiusHalfGap_density_ratio_continuousAt;
CMVBallCase.fixedRadiusHalfGap_density_ratio_tendsto_one_of_gt;
CMVBallCase.fixedRadiusHalfGap_density_ratio_tendsto_one_at_one;
CMVBallCase.fixedRadiusHalfGap_density_ratio_tendsto_one;
CMVBallCase.fixedRadiusHalfGap_pos_of_lt_radius_lt_one;
CMVBallCase.fixedRadiusHalfGap_pos_of_one_le_radius;
CMVBallCase.fixedRadiusHalfGap_pos_of_gt;
CMVBallCase.sameRadius_typeB_gt_typeC
```

Common perimeter and persistence

For $p > 2\lambda\pi$, let $r_B(p)$ and $r_C(p)$ be the unique positive radii satisfying

$$P_B(r_B(p)) = p, \quad P_C(r_C(p)) = p.$$

To see existence, note that $P_B(0) = 2\lambda\pi$, $P_C(\lambda) < 2\lambda\pi$, and

$$\pi R < P_C(R) < P_B(R) \quad (R > 0).$$

Continuity supplies the required radii, and strict monotonicity supplies uniqueness. The same-radius comparison then implies

$$r_B(p) < r_C(p):$$

otherwise monotonicity of P_C would give $p \leq P_C(r_B(p)) < p$. These inverses are used only on the genuine open common-perimeter domain.

LEAN REFERENCES

```
CMVBallCase.lean CMVBallCase.CommonPerimeter;
CMVBallCase.typeCPerimeterAtRadius_gt_pi_mul_radius;
CMVBallCase.existsUnique_typeBRRadius_at_perimeter;
CMVBallCase.existsUnique_typeCRRadius_at_perimeter;
CMVBallCase.typeBRRadiusAtPerimeter; CMVBallCase.typeCRRadiusAtPerimeter;
CMVBallCase.typeBRRadiusAtPerimeter_lt_typeCRRadiusAtPerimeter
```

Define the area difference at common perimeter by

$$G(p) = A_B(r_B(p)) - A_C(r_C(p)).$$

The perimeter derivatives are continuous and positive at positive radii, so the one-dimensional inverse function theorem applies locally. Using $A_B' = RP_B'$ and $A_C' = RP_C'$, the chain rule gives

$$\frac{d}{dp}A_B(r_B(p)) = r_B(p), \quad \frac{d}{dp}A_C(r_C(p)) = r_C(p).$$

Consequently

$$G'(p) = r_B(p) - r_C(p) < 0.$$

The mean value theorem on each compact subinterval of $(2\lambda\pi, \infty)$ makes G strictly decreasing throughout that domain. This is the monotone quantity used in the proof; no monotonicity of $I_B - I_C$ is assumed.

LEAN REFERENCES

```
CMVBallCase.lean CMVBallCase.commonPerimeterAreaGap;
CMVBallCase.typeBPerimeterAtRadius_hasStrictDerivAt;
CMVBallCase.typeCPerimeterAtRadius_hasStrictDerivAt;
CMVBallCase.commonPerimeterAreaGap_local_hasDerivAt;
CMVBallCase.commonPerimeterAreaGap_strictAnti
```

Now suppose $\lambda\pi < v_0 < v$ and $I_C(v_0) \leq I_B(v_0)$. Put $p_0 = I_B(v_0)$. At this perimeter the type-B area is v_0 . The type-C radius realizing area v_0 has perimeter at most p_0 , so monotonicity of its perimeter and area maps gives

$$A_C(r_C(p_0)) \geq v_0, \quad G(p_0) \leq 0.$$

Because I_B is strictly increasing, $p = I_B(v) > p_0$. Hence

$$G(p) < G(p_0) \leq 0, \quad A_C(r_C(p)) > v.$$

The unique type-C radius of area v is therefore strictly less than $r_C(p)$; its perimeter is strictly less than $p = I_B(v)$. This proves the theorem.

LEAN REFERENCES

```
CMVBallCase.lean CMVBallCase.commonPerimeterAreaGap_at_typeBProfile_nonpos;
                  CMVBallCase.typeCPerimeterAtArea_lt_of_commonPerimeterAreaGap_neg;
                  CMVBallCase.typeC_dominance_persists
```

Geometric scope

The verified result is persistence of type-C dominance over the faithful all-candidate type-B/C profiles. The formal argument establishes the full finite parameter domains, equality with the source formulas, existence and uniqueness at the required areas and perimeters, and the strict common-perimeter comparison. It does not itself establish existence or regularity of finite-perimeter minimizers, their connectedness, or the exhaustiveness of the geometric candidate classification. Nor does the persistence implication alone assert that a crossing exists.

The paper-level conclusion $v_1 = v_2$, proposed in the remark after [CMV, Theorem 3.23, p. 34], follows only when this analytic theorem is combined with the finite-perimeter existence, regularity, connectedness, and exhaustive candidate-classification premises underlying Proposition 3.22 and Theorem 3.23, together with the latter theorem’s small- and large-area alternatives. Under those external premises, above $\lambda\pi$ every minimizer is of type B or C, type B wins near that endpoint, and type C wins for sufficiently large areas. The continuous candidate profiles therefore meet; at their meeting both types are optimal, and persistence excludes any later return of type B. Thus the transition thresholds can be taken equal. These geometric-measure-theory premises and this global minimizer classification are not formalized in this package; they are distinct from the verified analytic profile theorem.

Bibliography

[CMV] Antonio Cañete, Michele Miranda Jr., and Davide Vittone, *Some isoperimetric problems in planes with density*, arXiv:0906.1256v1 [math.DG], 8 June 2009. Relevant locators: Proposition 3.22, pp. 32–33; Theorem 3.23 and its following remark, pp. 34–36; candidate formulas and Figure 19, pp. 34–35.